

MCA-416 Lab on Python Programming

1. Python Step-by-step Installation Process.

Step 1: Download Anaconda

Open the official Anaconda website:

<https://www.anaconda.com/>

Choose **Anaconda Distribution** for **Windows**.

Download the **Windows 64-bit Graphical Installer**.

Step 2: Start Installation

After downloading:

1. Open your **Downloads** folder.
2. Double-click the Anaconda installer .exe.
3. Click **Next**.

You will see the license agreement.

4. Select **I Agree**.
-

Step 3: Select Installation Type

You will get options:

- **Just Me (Recommended)**
- All Users

Select:

☒ **Just Me (Recommended)**

Click **Next**.

Step 4: Select Installation Location

Anaconda will suggest a location such as:

C:\Users\YourName\anaconda3

You can keep the default location.

Click:

Next.

Step 5: Advanced Installation Options

You may see options such as:

- ☐ Create shortcuts
 - ☐ Add Anaconda to my PATH environment variable
 - ☐ Register Anaconda as my default Python
 - ☐ Clear the package cache upon completion

For a beginner installation:

- ☒ **Create shortcuts**
- ☒ **Register Anaconda as my default Python**

Usually leave:

☒ **Add Anaconda to my PATH environment variable**

unchecked, because Anaconda generally recommends using Anaconda Navigator/Anaconda Prompt rather than adding Anaconda to PATH.

Click:

Install

Step 6: Wait for Installation

The installation may take several minutes.

Wait until you see:

Installation Complete

Then click:

Next → Finish

Step 7: Open Anaconda Navigator

Click the Windows **Start** button.

Search:

Anaconda Navigator

Open it.

You should see applications such as:

- Jupyter Notebook
- JupyterLab
- Spyder
- Anaconda Prompt

For your practicals, we can start with **Jupyter Notebook**.

Step 8: Start Jupyter Notebook

In Anaconda Navigator, find:

Jupyter Notebook

Click:

Launch

Your browser should open automatically.

You may see a page similar to:

Home
Files
Running

This means **Jupyter Notebook is working**.

2. Implement a program base on Strings.(slicing, casting, string function)

```
sentence = input("Enter a sentence: ")

# String slicing
first_word = sentence.split()[0] # Getting the first word
middle_section = sentence[5:15] # Getting a slice of the string from index 5 to 14
last_five_chars = sentence[-5:] # Getting the last 5 characters

# String casting
number_str = "123"
number = int(number_str) # Casting string to integer

# String functions
uppercase_sentence = sentence.upper() # Converting to uppercase
lowercase_sentence = sentence.lower() # Converting to lowercase
title_sentence = sentence.title() # Converting to title case
word_count = sentence.count(" ") + 1 # Counting words based on spaces
print("Original Sentence:", sentence)
print("First Word:", first_word)
print("Middle Section (5 to 14):", middle_section)
print("Last 5 Characters:", last_five_chars)
print("Casted Number:", number)
print("Uppercase Sentence:", uppercase_sentence)
print("Lowercase Sentence:", lowercase_sentence)
print("Title Case Sentence:", title_sentence)
print("Word Count:", word_count)
```

Output:

```
Enter a sentence: Python Programming
Original Sentence: Python Programming
First Word: Python
Middle Section (5 to 14): n Programm
Last 5 Characters: mming
Casted Number: 123
Uppercase Sentence: PYTHON PROGRAMMING
Lowercase Sentence: python programming
Title Case Sentence: Python Programming
Word Count: 2
```

3. Implement a program based on Simple statements using Variables, Built-in Data Types, Types of operator, etc.

```
def simple_statements_demo():
    # Variables and Built-in Data Types
    integer_var = 10    # integer
    float_var = 15.5    # float
    string_var = "Python" # string
    boolean_var = True  # boolean
    list_var = [1, 2, 3, 4, 5] # list
    tuple_var = (10, 20, 30) # tuple
    dict_var = {'name': 'John', 'age': 30} # dictionary

    # Print variables and their types
    print("Variables and their types:")
    print(f"Integer: {integer_var} (type: {type(integer_var)})")
    print(f"Float: {float_var} (type: {type(float_var)})")
    print(f"String: {string_var} (type: {type(string_var)})")
    print(f"Boolean: {boolean_var} (type: {type(boolean_var)})")
    print(f"List: {list_var} (type: {type(list_var)})")
    print(f"Tuple: {tuple_var} (type: {type(tuple_var)})")
    print(f"Dictionary: {dict_var} (type: {type(dict_var)})")

    # Arithmetic Operators
    print("\nArithmetic Operators:")
    print(f"{integer_var} + {float_var} = {integer_var + float_var}")
    print(f"{integer_var} - {float_var} = {integer_var - float_var}")
    print(f"{integer_var} * {float_var} = {integer_var * float_var}")
    print(f"{integer_var} / {float_var} = {integer_var / float_var}")
    print(f"{integer_var} % 3 = {integer_var % 3}")
    print(f"{integer_var} ** 2 = {integer_var ** 2}")
    print(f"{integer_var} // 3 = {integer_var // 3}")

    # Comparison Operators
    print("\nComparison Operators:")
    print(f"{integer_var} == 10: {integer_var == 10}")
    print(f"{integer_var} != 10: {integer_var != 10}")
    print(f"{integer_var} > 5: {integer_var > 5}")
    print(f"{integer_var} < 15: {integer_var < 15}")
    print(f"{integer_var} >= 10: {integer_var >= 10}")
    print(f"{integer_var} <= 10: {integer_var <= 10}")

    # Logical Operators
    print("\nLogical Operators:")
    print(f"True and False: {True and False}")
    print(f"True or False: {True or False}")
    print(f"not True: {not True}")

    # Assignment Operators
    print("\nAssignment Operators:")
```

```

x = 5
print(f"x = 5: {x}")

x += 3
print(f"x += 3: {x}")

x -= 2
print(f"x -= 2: {x}")

x *= 2
print(f"x *= 2: {x}")

x /= 3
print(f"x /= 3: {x}")

x %= 2
print(f"x %= 2: {x}")

x **= 3
print(f"x **= 3: {x}")

x //= 2
print(f"x //= 2: {x}")

if __name__ == "__main__":
    simple_statements_demo()

```

Output:

Variables and their types:

```

Integer: 10 (type: <class 'int'>)
Float: 15.5 (type: <class 'float'>)
String: Python (type: <class 'str'>)
Boolean: True (type: <class 'bool'>)
List: [1, 2, 3, 4, 5] (type: <class 'list'>)
Tuple: (10, 20, 30) (type: <class 'tuple'>)
Dictionary: {'name': 'John', 'age': 30} (type: <class 'dict'>)

```

Arithmetic Operators:

```

10 + 15.5 = 25.5
10 - 15.5 = -5.5
10 * 15.5 = 155.0
10 / 15.5 = 0.6451612903225806
10 % 3 = 1
10 ** 2 = 100
10 // 3 = 3

```

Comparison Operators:

```

10 == 10: True
10 != 10: False
10 > 5: True
10 < 15: True
10 >= 10: True
10 <= 10: True

```

Logical Operators:

```
True and False: False
True or False: True
not True: False
```

Assignment Operators:

```
x = 5: 5
x += 3: 8
x -= 2: 6
x *= 2: 12
x /= 3: 4.0
x %= 2: 0.0
x **= 3: 0.0
x //= 2: 0.0
```

Other Program:

Variables and Built-in Data Types

```
name = "John"      # String
age = 25           # Integer
height = 5.9       # Float
is_student = True  # Boolean
grades = [85, 90, 88] # List (array-like data type)
address = None     # NoneType (for uninitialized variables)
```

Printing variables

```
print("Name:", name)
print("Age:", age)
print("Height:", height) print("Is
Student:", is_student)
print("Grades:", grades)
print("Address:", address)
```

Basic Operators

Arithmetic Operators

```
addition = age + 5      # Addition
subtraction = age - 3   # Subtraction
multiplication = age * 2 # Multiplication
division = age / 2      # Division (returns float)
integer_division = age // 2 # Integer division (returns int)
modulus = age % 2       # Modulus (remainder) exponent =
age ** 2                # Exponentiation (age^2)
```

Comparison Operators

```
is_adult = age >= 18
is_minior = age < 18
```

Logical Operators

```
is_eligible_for_discount = is_student or age < 18
```

String Operators

```
full_name = name + " Doe" # String concatenation
repeated_name = name * 3   # String repetition
```

List Operations

```
grades.append(95) # Adding a grade to the list
average_grade = sum(grades) / len(grades) # Calculate average grade
```

```

# Output the results
print("\nArithmetic Operations:")
print("Addition (age + 5):", addition)
print("Subtraction (age - 3):", subtraction)
print("Multiplication (age * 2):", multiplication)
print("Division (age / 2):", division)
print("Integer Division (age // 2):", integer_division)
print("Modulus (age % 2):", modulus)
print("Exponent (age ** 2):", exponent)

print("\nComparison & Logical Operations:")
print("Is Adult (age >= 18):", is_adult)

print("Is Minor (age < 18):", is_minor)
print("Is Eligible for Discount:", is_eligible_for_discount)

print("\nString Operations:") print("Full
Name:", full_name)
print("Repeated Name:", repeated_name)

print("\nList Operations:")

print("Updated Grades:", grades)
print("Average Grade:", average_grade)
Output:
Name: John
Age: 25
Height: 5.9
Is Student: True
Grades: [85, 90, 88]
Address: None

Arithmetic Operations:
Addition (age + 5): 30
Subtraction (age - 3): 22
Multiplication (age * 2): 50
Division (age / 2): 12.5
Integer Division (age // 2): 12
Modulus (age % 2): 1
Exponent (age ** 2): 625

Comparison & Logical Operations:
Is Adult (age >= 18): True
Is Minor (age < 18): False
Is Eligible for Discount: True

String Operations:
Full Name: John Doe
Repeated Name:
JohnJohnJohn

List Operations:

Updated Grades: [85, 90, 88, 95]
Average Grade: 89.5

```

4. Implement a program based on decision and looping statements

```
i = 0

# The loop breaks when it reaches 'h'
for letter in 'Python':
    if letter == 'h':
        print('Yes, h is present in string')
        break
    else:
        print('Current Letter:', letter)

var = 10

while var > 0:
    print('Current variable value:', var)
    var = var - 1

    if var == 5:
        break

print('Good bye!')
```

Output:

```
Current Letter: P
Current Letter: y
Current Letter: t
Yes, h is present in string
Current variable value: 10
Current variable value: 9
Current variable value: 8
Current variable value: 7
Current variable value: 6
Good bye!
```